4.2

We approximate this binomial distribution ($X \sim Bin(n = 1000, p = 0.42)$) because $np(1-p) > 10$.

$E(X) = np = 420$

$Var(x) = np(1-p) = 159.6$

$X \sim N(420, 159.6)$

$\sigma = \sqrt{159.6} = 12.633$

$450 = 420 + 12.633z \rightarrow 30 = 12.633z \rightarrow z = 2.37$

$P(X > 450) = 1 - P(X < 450) = 1 - \Phi(2.37) = 1 - 0.9972 = 0.0028$

4.6

$2\Phi(2\epsilon\sqrt{n}) \geq 0.95 \rightarrow \Phi(2\epsilon\sqrt{n}) \geq 0.975$

$\rightarrow 2\epsilon\sqrt{n} \geq 1.96 \rightarrow n = \frac{1.96}{2\epsilon}^2$

$\epsilon = 0.02$ from the problem

$n = \frac{1.96}{0.04}^2 = 2401$

4.10

$1 - P(X = 0) = 0.5 \rightarrow P(X = 0) = 0.5$

$P(X = 0) = e^{-\lambda} * 1 = 0.5 \rightarrow \lambda = -\ln(0.5) = 0.693147$

Scoring exactly 3 goals (hat trick) is then $P(X = 3) = 0.5\frac{3^{-\ln(0.5)}}{3!} = 0.178457$

If you define hat trick as 3 or more goals then its $1 - P(X = 0) - P(X = 1) - P(X = 2)$ but since the majority accepted definition is exactly 3 goals I will leave my answer as $P(HATTRICK) = P(X = 3) = 0.178457$

4.16

a

$P(Start1) = \frac{1.99999999-1.5}{4.8-1.5} \approx \frac{0.5}{3.3} = \frac{5}{33}$

$X \sim Bin(500, \frac{5}{33})$

$X \sim N(\frac{2500}{33}, \frac{70000}{1089})$

b

$P(Start3) = \frac{3.99999999-3}{4.8-1.5} \approx \frac{1}{3.3} = \frac{10}{33}$

$X \sim Bin(500, \frac{10}{33})$

$X \sim N(\frac{5000}{33}, \frac{115000}{1089})$

4.24

4.26

4.35